PRINCIPAL DIVISORS GENERATE RATIONAL EQUIVALENCE ON A VARIETY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 02RQ: sources/stacks-chow/tags.txt, source lines 3633–3633. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02RT: sources/stacks-chow/tags.txt, source lines 3636–3636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a variety. Let $W \subset X$ be a subvariety of dimension $k+1$. Let $f \in \mathbf{C}(W)^*$ be a nonzero rational function on W. Then $\mathrm{div}_W(f)$ is rationally equivalent to zero on X. Conversely, these principal divisors generate the abelian group of cycles rationally equivalent to zero on X.*

Proof. The first assertion follows from Chow Homology, Lemma 02RQ. More precisely, let $W' \subset X \times \mathbf{P}^1$ be the closure of the graph of f. Then $\mathrm{div}_W(f) = [W'_0]_k - [W'_\infty]$ in $Z_k(W) \subset Z_k(X)$, see part (6) of Chow Homology, Lemma 02RQ.

For the second, let $W' \subset X \times \mathbf{P}^1$ be a closed subvariety of dimension $k+1$ which dominates $\mathbf{P}^1$. We will show that $[W'_0]_k - [W'_\infty]_k$ is a principal divisor which will finish the proof. Let $W \subset X$ be the image of W' under the projection to X. Then $W \subset X$ is a closed subvariety and $W' \to W$ is proper and dominant with fibres of dimension 0 or 1. If $\dim(W) = k$, then $W' = W \times \mathbf{P}^1$ and we see that $[W'_0]_k - [W'_\infty]_k = [W] - [W] = 0$. If $\dim(W) = k+1$, then $W' \to W$ is generically finite[1]. Let f denote the projection $W' \to \mathbf{P}^1$ viewed as an element of $\mathbf{C}(W')^*$. Let $g = \mathrm{Nm}(f) \in \mathbf{C}(W)^*$ be the norm. By Chow Homology, Lemma 02RT we have

$$\mathrm{div}_W(g) = \mathrm{pr}_{X,*}\mathrm{div}_{W'}(f)$$

Since $\mathrm{div}_{W'}(f) = [W'_0]_k - [W'_\infty]_k$ by Chow Homology, Lemma 02RQ the proof is complete. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

[1] If $W' \to W$ is birational, then the result follows from Chow Homology, Lemma 02RQ. Our task is to show that even if $W' \to W$ has degree > 1 the basic rational equivalence $[W'_0]_k \sim_{rat} [W'_\infty]_k$ comes from a principal divisor on a subvariety of X.